

# MeiMeiRoRo

Maze VR experience with nested structure and body tilting creating two consciousnesses

Osuke Funabiki  
243426031@ccmailg.meijo-u.ac.jp  
Meijo University  
Nagoya, Japan

Naoki Morita  
243426038@ccmailg.meijo-u.ac.jp  
Meijo University  
Nagoya, Japan

Yasuyuki Yanagida  
yanagida@meijo-u.ac.jp  
Meijo University  
Nagoya, Japan

## Abstract

“MeiMeiRoRo” is an interactive experience in a nested maze, and its concept is to present two consciousnesses, active and passive. In this experience, operations to a miniature avatar, such as tilting, are returned to the user through a nested structure. Operations on the miniature avatar are an active awareness, and when the resulting operations are returned to the user, a passive awareness occurs. These two consciousnesses are the concept, which works because the nested structure implies that the operations on the miniature avatar are an operation on him/herself.

## CCS Concepts

• **Human-centered computing** → **Heuristic evaluations; Interaction design theory, concepts and paradigms.**

## Keywords

interaction, nested, tilt, platform

## ACM Reference Format:

Osuke Funabiki, Naoki Morita, and Yasuyuki Yanagida. 2024. MeiMeiRoRo: Maze VR experience with nested structure and body tilting creating two consciousnesses. In *SIGGRAPH Asia 2024 XR (SA XR '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3681759.3688928>

## 1 Introduction

Nested structures, represented by Matryoshka dolls, have been used as an interface to make object manipulation in VR more explicit [Stoakley et al. 1995] and a game concept to create a unique worldview. In addition, there has been much research on locomotion interfaces that allow free walking in VR [Darken et al. 1997] and experiences that induce a sense of inclination [Lee and Lishman 1975]. On the other hand, the majority of experiences using nested structures are expressed only through visual stimulation, and few interaction systems that combine body tilting let users experience the unique world of the nested structure. Therefore, we created “MeiMeiRoRo” as an innovative experience that also presents a body tilt, taking advantage of the characteristics of the nested structure. Figure 1 shows the appearance of the device, with a tiltable boarding surface. The “MeiMeiRoRo” presents an experience in which

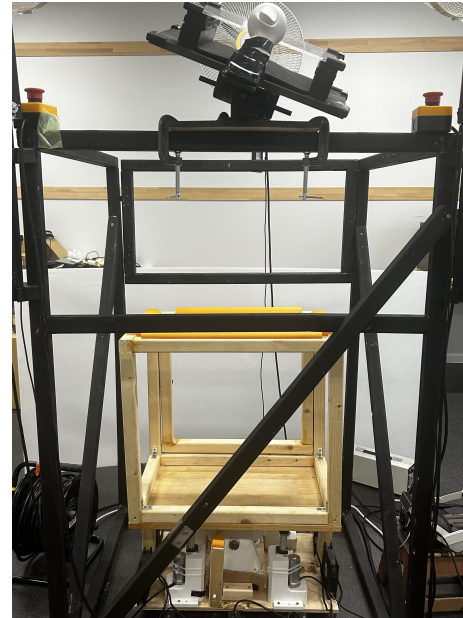
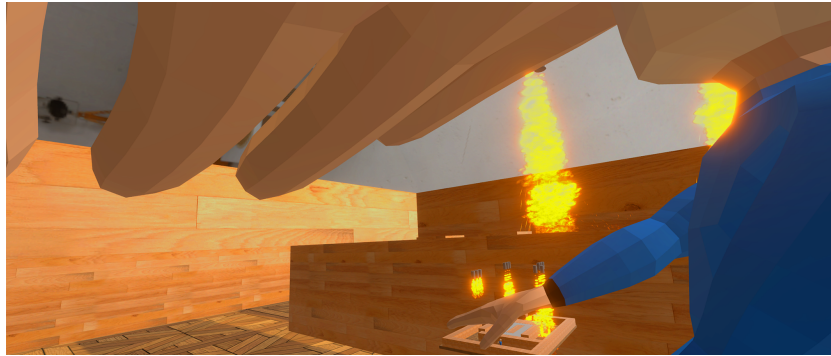


Figure 1: Device Overview

the tilt that the user gives to the miniature is also given to the user's actual body through the nested structure. Involving not only visual stimulation but also the body's inclination, nested worlds are presented as an interaction that pursues more reality.

## 2 Experience Overview

The nested maze world presents an experience in which the user feels two consciousnesses, active and passive, arising within the user. They enter a huge maze world and try to reach their goal. They hold a maze board in their hands, and this maze is the same as the huge maze world they are in. Therefore, what happens on the maze board in front of them also happens in the huge maze world they are in, and this is the effect of a nested structure. They can see a small user inside the maze board from an overhead view, and by tilting and sliding the small user inside the board, the user can also move around inside the huge maze. In this instance, the user tilts the maze board with active awareness, sliding him/herself in the direction they want to move. On the other hand, the action of tilting the maze board is reflected in the world of the huge maze, and their own body is tilted similarly. As a result, the user has a passive awareness that the huge maze is tilted and the user is slid. Thus, “tilting the maze board” makes the user rise to both active and passive consciousness. This concept is possible because the



**Figure 2: A fire gimmick that prevents the user**

experience is nested and involves inclination. The maze has some gimmicks that prevent the user. These gimmicks are important elements in making the experience enjoyable, and each one symbolizes the nesting structure. Figure 2 is one of the gimmicks.

For ease of understanding, Figure 2 shows the user as seen from the left side, but the actual experience proceeds from their perspective on the right side of the figure. In the figure, flames are thrown toward the user from above in the huge maze world, and the same gimmick is on the maze board held in the user's hands. In this scene, they block the flames on the maze board with the user's hand. This interference is reflected in the huge maze world through its nested structure, appearing as a huge hand protecting him/herself from the flames (the upper side of Figure 2). To summarize, the nested structure in this experience refers to the linked huge maze and the maze board in the user's hands, and this mechanism, combined with actual body tilt, realizes the arising of two consciousnesses, active and passive.

### 3 Device Detail

The "MeiMeiRoRo" consists of two main devices: a maze controller, a tilt device and safety devices. A detailed explanation of each device is provided below.

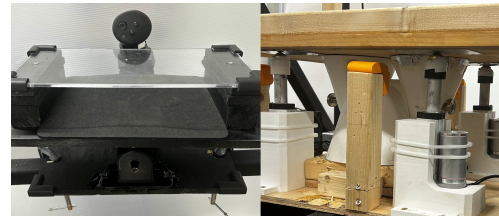
#### 3.1 Maze Controller

It functions as a device for an operation that is grasped with both hands during the experience. Figure 3 left side shows the maze controller. During the experience, a maze board is superimposed on the position of the maze controller, and the user moves through the maze by tilting the board. The maze controller has two orthogonal rotation axes on its base to achieve 2-DOF rotation in roll and pitch. A controller (meta quest touch plus) is attached to the maze controller, and the tilt is acquired by the built-in IMU.

#### 3.2 Tilt Device

It functions as the floor surface of the huge maze on which the users are standing. The device tilts similarly, reflecting the tilt of the maze controller. Figure 3 right side shows the tilt device. The device consists of a cube-shaped standing surface, tilt axes, and linear actuators that support the trolley from below. The tilt axis is the main component that supports the load of the user, and it provides 2-DOF rotation in roll and pitch. Since the device tilts in accordance

with the tilt of the maze board, the mechanism is necessarily the same as the maze controller, but it is designed to be durable and large enough to support the user's load. Tilting is achieved by linear actuators that expand and contract. The tilt amount of the maze controller is recalculated into the actuator extension/retraction amount in the tilt device, and the appropriate extension/retraction operation is performed to reproduce the intended tilt amount.



**Figure 3: Controller(left) and Tilt Device(right)**

#### 3.3 Safety Device

The design takes sufficient care not to endanger the users or the surroundings because they are standing on the tilt device. The cabinet is equipped with preventive and operational safety measures. As a precautionary measure, a fence is built to enclose the users. It functions to assist them in getting in and out of the device or when they lose their posture during the experience. As an operational measure, an emergency stop button is equipped. The exhibitor activates it when a danger or abnormality occurs during the experience. When the emergency stop is activated, the power supply is cut off and the physical movement of the entire cabinet, including the actuators, stops.

### References

- Rudolph Poul Darken, William R Cockayne, and David Carmein. 1997. The omnidirectional treadmill: a locomotion device for virtual worlds. *UIST97: Tenth Annual Symposium on User Interface Software and Technology* (Oct. 1997). <https://doi.org/10.1145/263407.263550>
- David N. Lee and J.R. Lishman. 1975. Visual proprioceptive control of stance. *Journal of Human Movement Studies* 1, 1 (1975), 87–95.
- Richard Stoakley, Matthew J. Conway, and Randy Pausch. 1995. Virtual Reality on a WIM: Interactive Worlds in Miniature. *CHI95: Conference on Human Factor in Computing Systems* (May 1995), 265–272. <https://doi.org/10.1145/223904.223938>